

Bringing Emotions into Practice: A Framework for AI Design to Support Emotion work

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ABSTRACT

Informal or family caregivers receive no professional training and are overburdened. The CSCW community has increasingly examined ways to support informal caregivers. However, the solutions have primarily focused on managing the negative emotions of informal caregivers. Informal caregivers go through a range of emotional experiences, which are often invisible to others. In this work, we aim to use the context of informal caregiving needs and experiences to understand how we can design future Artificial Intelligence (AI) systems for supporting emotion work. We conducted user studies and interviews to understand what informal caregivers perceive as emotion work in two caregiving contexts: parenting and Alzheimer's Disease and Related Dementias (ADRD). Based on the findings, we propose a framework for generating design requirements for AI systems that support emotion work. Our work contributes to how we can design AI to support people's practices that involve emotion work.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Empirical studies in HCI**.

KEYWORDS

Emotion work; Artificial Intelligence; Informal Caregivers

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1 INTRODUCTION

Informal caregivers include family, friends, or other personal acquaintances, who tend to “live to provide care” to the care recipient [32]. Informal caregiving is not simply an imposed duty but arises from emotional bonds with the care recipient and is usually assumed without formal preparation or monetary remuneration [12]. These caregivers are often overburdened. Caregiving burden

can lead to higher levels of anxiety, depression, sleep problems, and chronic fatigue [4]. CSCW researchers have recognized such challenges of informal caregivers and have called for attention to supporting caregiver burden, given that much attention has been done so far in supporting care recipients [6]. Given the high psychological challenge of caregivers, many focused on improving the negative psychological outcomes of informal caregivers, such as depression, anxiety, and strain [2, 15, 28, 30, 36] and what Artificial Intelligence (AI) systems can do to off-load the caregiving burden [11, 17, 20, 22, 37].

However, these AI solutions are not always effective, with the developed solutions not fully capturing informal caregivers' experiences and subjective burden in caregiving [9, 10, 14, 28, 34]. Informal caregivers face varied emotional challenges as they try to balance managing their emotions while supporting the care recipient [23, 33]. Recent studies report that there are also positive experiences of being an informal caregiver [1, 7, 24]. This spectrum of emotions that informal caregivers experience and tend to, their own as well as those of others, forms **emotion work** [18, 23, 33]. CSCW literature in informal caregiving has developed novel and creative solutions on how caregiving burden and negative emotional experiences can be supported. Unpacking the varied emotional experiences of caregiving that make up emotion work and how they play a role in designing for informal caregivers is still underexplored. To better understand the complete emotional experiences of informal caregivers and look for possible ways to help them, more work is needed to understand the emotion work of caregivers and how AI systems can support it.

The research questions that we aim to understand are: **RQ1**. What kind of emotion work is involved in informal caregiving? **RQ2**. How should we guide ourselves to design AI systems that can support the emotion work of informal caregiving and beyond?

The contribution of this proposed work is twofold. First, we aim to understand and articulate the complexities of emotion work in informal caregiving. The emotion work of informal caregivers needs to be made visible in ways that can be understood for solutions to be developed for them [29]. Second, we aim to provide a framework for mapping the emotion work of informal caregiving and suggest guidelines for designing AI systems that can help with the underexplored emotion work. Future goals would be to inform how the framework can be extended to other contexts that require emotion work, such as long-distance relationships, sibling relationships, or mentoring. Given the strong emotion work informal caregivers experience, informal caregiving provides an effective starting point for developing this ambitious plan.

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2 CONCEPTUAL LENS: THE THEORY OF EMOTION PRACTICE

Emotion practice theory demonstrates how emotion is inherently social as its management and illustrates why emotions should be theorized and studied as components of work and occupations, including caregiving [8, 35]. An emotion practice theoretical framework uses concepts from Bourdieu's practice theory (i.e., field, capital, and habitus) [3, 8]. **Field** explains the social and physical context of emotions, including the different actors involved. **Capital** emphasizes the skills necessary to manage these emotions. **Habitus** shows the person's internal motivation to respond to a context and develop their skills further and how this motivation is dependent on social factors. Using the field, capital, and habitus, the emotion practice theory also provides a way to conceptualize emotion as a feature of social practice, comprising of parts that can be helped with or supported.

This approach can help CSCW researchers capture the complexity of emotions within the ever-changing social and physical contexts of informal caregiving [13]. In this work-in-progress, we introduce using the emotion practice theory to understand the complexity of emotions in informal caregiving. Using this theory, we develop a starting point for building a framework to guide the design of AI systems for supporting emotion work.

3 PRELIMINARY STUDIES

Our work is situated in two informal caregiving contexts: parenting and ADRD informal caregiving. Caregiver-care recipient relationship is important in these contexts and shapes the quality of care [16, 25]. We conducted preliminary user studies and interviews studies with parents and ADRD caregivers to understand what made up emotion work in these contexts (RQ1).

3.1 Study 1: User Study of a Conversational Agent (CA) for Parenting

The first study aimed to understand the perceptions of a CA developed for eating healthy in the context of parenting. We conducted user test sessions of the CA prototype to understand its perceived acceptability and usefulness by 24 parents for developing healthy eating habits in themselves as well as their children [26]. After parents' interaction with the CA prototype in each user test session, we used qualitative interviews to understand parents' perceptions and suggestions for improvements in the prototype. We transcribed the interviews and conducted a thematic analysis [5]. Findings showed parents' perceived usefulness of the CA, leading to implications for conversational AI systems. We also found some parenting practices that parents did not want the AI system to help them with their children, such as suggesting food items for and having conversations around healthy eating with their children. Parents felt they were emotionally connected to these tasks— they were responsible for shaping their children's habits from an early age and wanted to be involved in helping their children make decisions. Such apprehensiveness around the use of AI systems raised essential questions about what caregiving work AI systems should help caregivers with, and how their emotional connection with these tasks affected their perception of receiving help from AI systems.

"Parents can help children obtain really good habits when they're young. I [parent] used to have really good habits because **my mom would teach me** about the kind of food groups and how those things react. It's those changes in my adult life that have gotten me to where I am. So, I would think that would be a great way for kids to get more information and make better choices for themselves." -Parent

3.2 Study 2: Interview Study with ADRD Caregivers and Parents

The second study was to understand in-depth the subjective caregiver burden. That is, to generate a deeper understanding of the variety and intensity of emotions associated with caregiving tasks. We interviewed 15 informal caregivers (12 ADRD caregivers and 3 parents) to understand their needs and current coping strategies for overcoming the caregiving burden. We transcribed the interviews and conducted a thematic analysis [5]. Findings showed that the caregivers associated different personal meanings with caregiving. They wanted AI systems to take over some of the everyday, utilitarian caregiving tasks. At the same time, they did not want AI systems to take over the tasks that made them feel emotionally connected with the care recipient. A mix of positive and negative emotions was involved, with their intensities changing with tasks caregivers considered essential and as they progressed through their caregiving journey. These tasks were where caregivers required the most help, and current AI systems failed to provide relevant help. These findings called for a better understanding of these tasks that make up the emotion work of informal caregiving so that future AI systems can be better designed to provide support as needed by the caregivers.

"I think the task I would not want to hand off would be helping him [PLWD] to take his bath in the morning. I always loved to do that because I remember when I was young, too, he wanted to help me with my baths. **I would still want to help him with his bath. That's like trying to cherish him more.**" - ADRD caregiver

4 ONGOING WORK ON THE FRAMEWORK DEVELOPMENT

The preliminary studies resulted in understanding and helping to identify emotion work in informal caregiving. To elaborate on potential opportunities for AI systems to help with emotion work, we are developing a framework for generating design requirements for designing AI systems that support emotion work (RQ2).

The framework has two goals: (a) strategic and (b) practical. The strategic goal of the framework is to help researchers identify and map emotion work onto the framework to provide a landscape of the work that AI systems can support. The practical goal of the framework is to help researchers derive requirements on how emotion work can be supported through AI. Below, we exemplify using the framework in parenting and ADRD caregiving contexts.

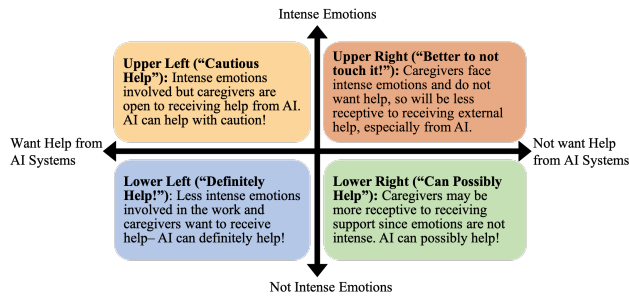


Figure 1: The Proposed Framework with Two Dimensions: Intensity of Emotions Involved and Caregivers Wanting Help from AI Systems.

4.1 Strategic Goal

Based on the insights from the preliminary studies, we found that emotion work varied along the dimensions of (1) the intensity of emotions involved and (2) to what extent caregivers wanted support from AI systems. These two dimensions form the basis of the framework, as shown in Fig 1. Steps 1 and 2 below help us achieve this goal.

- **Step 1.** The first step would be to identify and map the emotion work from the preliminary studies onto the framework. As we found from the preliminary studies, emotion work is work that informal caregivers undertook to: (1) juggle the multitude of emotions that come with caregiving, (2) maintain a positive experience in the changing nature of caregiving, (3) and manage intense emotions to not make it harmful for themselves or anyone involved. Such emotion work was often tied to the caregivers' identity and value as a caregiving person (e.g., 'feeling honored'). It was also tied with all kinds of tasks that one might want others to do for them (e.g., washing hair, cooking). The emotion work identified varied along with the intensity of emotions involved and whether help from AI was needed and could map onto one of the four quadrants as shown in Fig 2. For example, parents wanted help from AI systems for initiating conversations around eating healthy with their children while giving them the freedom to plan it, which can be mapped to the Lower Left quadrant in the framework. AD RD caregivers did not want help resolving arguments with care recipients, wanting to manage the intense emotions themselves, which can be mapped to the Upper Right quadrant in the framework. The four quadrants help us identify what emotion work can be feasibly supported by AI systems and what work AI systems should not support.
- **Step 2.** Caregivers from the preliminary studies did not want to hand off emotion work completely to AI systems but only wanted support for some parts of this work. Since emotions are not always rational and objective, the theory helps to understand the parts of emotion work that are objective and easy to support by AI (field, capital) versus parts that are subjective and difficult to support by AI (habitus). For instance, in the case where AD RD caregivers did not want help

from AI systems for resolving arguments with care recipients, the *field* included the actors (caregivers and the care recipient), the location where the emotion work happened (their homes), and the context (delay in helping to use the bathroom). The *capital* included the skills caregivers needed to develop to manage that situation, for example, calming the care recipient down while keeping calm themselves. The field and capital parts are objective and easier to design AI solutions for. The *habitus* included the motivation of caregivers to prioritize this emotion work because they have dealt with this situation previously and understand that it is essential for the care recipient or because they think it is crucial for their relationship building with the care recipient. The habitus part is more subjective to caregivers, and generating design solutions without incorporating caregivers' perspectives is challenging. We look at designing AI for the objective parts as it is achievable through the following steps. For the subjective part, future work will involve a participatory design [27] study to generate ideas together with the caregivers.

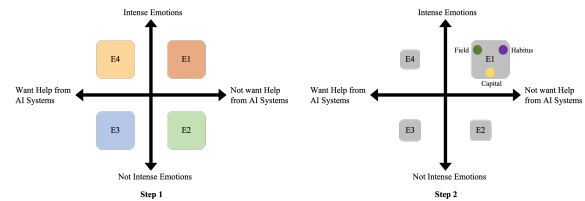


Figure 2: Strategic Goal of the Framework. Step 1: Identify and map the emotion work (E1-E4) onto the framework. Step 2: Using the Emotion Practice Theory, get the field, capital, and habitus parts of E1-E4.

4.2 Practical Goal

After identifying the areas of emotion work that would likely require external support, we now bring these findings to the practical goal of the framework—to provide requirements for designing AI systems to support emotion work in informal caregiving. Steps 3, 4, and 5 help to achieve this goal.

- **Step 3.** The third step would be to conduct a literature review of AI interventions, systems, solutions, or design components developed to date to support informal caregivers in the two contexts of parenting and AD RD caregiving. Next, the solutions from the literature review will be mapped to the emotion work in the framework in Step 1 to help understand which quadrants of emotion work have been addressed so far. For example, Moharana et al. [21] proposed a robot to facilitate communication between AD RD caregivers and care recipients and redirect the care recipients when distressed. This solution can be mapped to the Upper Left quadrant with this emotion work. This process is shown in Fig 3.
- **Step 4.** The fourth step would be to use affinity mapping [19] to group the mapped design solutions with the parts of the emotion practice theory for the emotion work the solutions

help with. Using the example from Step 3, we will group the design solution (the robot) to the emotion work's (managing care recipients' emotions when they are distressed) field and capital parts, as the robot helps caregivers develop communication skills in the context when the care recipient is distressed. However, the robot doesn't cater to the motivation of the caregiver to prioritize this work over others (habitus). This analysis will help us understand the gaps in the literature— which parts of the emotion work the design solutions have addressed and which are still unexplored. Using the design methodology of brainstorming and ideation with other HCI researchers, the next step would be to generate design requirements for the unexplored, objective parts (field and capital) of emotion work that can be feasibly met through AI systems. We will conduct this step for each context of parenting and ADRD caregiving and then find solutions common and different for the two contexts. Future work will also include conducting participatory design [27] sessions to generate design AI solutions for the subjective parts (habitus) together with the caregivers.

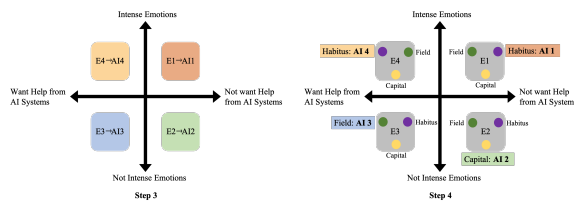


Figure 3: Practical Goal of the Framework. Step 3: Map the AI design solutions (A1-A4) from the literature review to the emotion work in the framework. Step 4: Group A1-A4 to the field, capital, or habitus parts of E1-E4 they are designed for.

- **Step 5.** The fifth step would be to evaluate the design requirements from the framework with the stakeholders for its validity. For instance, in our work, we plan to interview informal caregivers to understand their opinions on the design requirements generated from the framework. We will recruit parents and informal ADRD caregivers until we reach theoretical data saturation for each participant segment [31]. We will first instruct the participants on the meaning of the dimensions and the three parts derived using the Emotion Practice Theory. We will then present them with the design requirements for AI systems generated from the framework and ask them to confirm whether the classification of the requirements is correct. We will also ask them what they liked/did not like about the design requirements and probe the participants on how they would want to change or improve the requirements. The findings will help understand the user-confirmed validity of the mapping results in the framework and how it can be improved to incorporate the informal caregivers' perspectives.

5 DISCUSSION

The emotion work of informal caregivers is difficult to articulate due to its complex nature or not being visible [18, 29]. Prior CSCW

work supported mostly negative [2, 30] and sometimes positive [24] emotions of informal caregivers. We build on this body of work by proposing a framework that will serve as a tool for researchers to articulate and take the first step towards designing AI for the emotion work of informal caregivers comprising of varied emotions. Informal caregivers in our studies were apprehensive about AI taking over their roles as caregivers, and reducing their bond with the care recipient. The proposed framework allows for the design of AI systems to support, and not replace, the emotion work of informal caregivers. Involving informal caregivers in the framework development process ensures that their perspectives are taken into account. We believe the framework can help AI designers to augment social relationships and can extend to domains that require emotion work. Examples of such domains are long-distance relationships, sibling relationships, friendships, or even mentoring.

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